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When RLVR Shrinks the Reasoning Boundary: Diagnosing Pass@k Inversion

Reinforcement Learning with Verifiable Rewards Boosts One-Shot Accuracy While Silently Collapsing Sampling Diversity—a Failure Mode Traced to Boundary Prompts Where Rare Correct Trajectories Vanish Before They Can Be Reinforced

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Reinforcement learning with verifiable rewards (RLVR) has become the dominant recipe for training reasoning models, and its track record at improving pass@1 is well-documented. A new preprint reveals the other side of the ledger: RLVR can simultaneously *decrease* pass@k for large k , meaning the trained model solves fewer distinct problems under repeated sampling than the base model it was fine-tuned from. The author terms this phenomenon **pass@k inversion** and traces it to a concrete failure mode involving rare correct trajectories.

AT A GLANCE

Problem: RLVR is evaluated almost exclusively at pass@1, masking the fact that it can reduce coverage under diverse sampling.

Why it matters: Any system using best-of- N , voting, or repeated sampling to boost reliability may be *harmed* by RLVR training on boundary prompts.

Method: Pass@k evaluated pre- and post-RLVR across MATH and GSM8K; boundary prompts identified by base model entropy; causal rollout analysis tracing the loss of rare correct trajectories.

Result: RLVR improves pass@1 by 5–12 pp while reducing pass@16 by 3–8 pp on 15–25% of test problems.

Evidence: Per-Problem Base Anchoring (PBA) recovers 60% of the pass@k loss while preserving 95% of the pass@1 gains.

The Big Picture

Pass@k is a standard metric for reasoning benchmarks: it asks what fraction of problems a model can solve if given k independent sampling attempts, and a correct answer in any one attempt counts. For a fixed pass@1 score, higher pass@k reflects richer exploration of the solution space—a property valuable when models are used

with sampling-based inference strategies such as best-of- N selection, majority voting, or Monte Carlo Tree Search.

The intuition is that if RLVR improves a model, it should improve both pass@1 and pass@k. This paper shows the intuition is wrong for a non-trivial fraction of problems. After RLVR, the policy is sharper—it assigns higher probability to a narrower set of trajectories—and this sharpening can eliminate rare-but-correct paths that the base model would occasionally find at large k .

Why Existing Approaches Fall Short

Standard RLVR training (e.g., GRPO) collects G rollouts per problem, assigns binary rewards, and updates the policy to increase the relative likelihood of correct rollouts over incorrect ones. This mechanism has an implicit vulnerability: if none of the G rollouts for a problem are correct, the update treats the problem as definitively unsolvable and suppresses all trajectories toward its vicinity in policy space.

For **boundary prompts**—problems where the base model’s correct-trajectory probability is, say, 5–15%—a rollout group of size $G = 8$ will observe zero correct rollouts in a large fraction of training encounters. Each such encounter applies a gradient that reduces the probability of rare correct trajectories, eventually driving them to near zero even though they were genuinely present.

Core Mechanism

The author proposes a *two-mode account* of post-RLVR policy distributions:

- **Safe prompts:** Base model has high enough correct-trajectory probability that at least one correct rollout appears reliably in each group. RLVR reinforces and concentrates this probability. Pass@k improves monotonically with pass@1.
- **Boundary prompts:** Correct trajectories are too sparse to survive finite rollout groups. RLVR receives an absence-of-evidence signal and drives correct-trajectory probability toward zero. Pass@k *collapses* even as pass@1 may improve on adjacent safe prompts in the same batch.

Formally, for a boundary prompt x with base policy correct probability $\pi_{\text{base}}(x) \in [\epsilon, \delta]$ (small $\epsilon, \delta > 0$), the expected number of correct rollouts in a group of size G is $G \cdot \pi_{\text{base}}(x)$. When this falls below 1, the probability of observing zero correct rollouts in the group is approximately $(1 - \pi_{\text{base}})^G$, which for $\pi_{\text{base}} = 0.08$ and $G = 8$ is 50.6%.

Technical Formulation

Define pass@k for a single problem x and policy π as:

$$\text{pass@}k(x, \pi) = 1 - (1 - \pi_{\text{correct}}(x))^k$$

where $\pi_{\text{correct}}(x) = \sum_{a \in \mathcal{A}(x)} \pi(a | x)$ is the policy’s total mass on correct answers. Pass@ k inversion on problem x occurs when:

$$\pi_{\text{RL,correct}}(x) < \pi_{\text{base,correct}}(x)$$

even if $\pi_{\text{RL,correct}}(x) > \pi_{\text{base,correct}}(x)$ holds *on average* across all problems.

The GRPO update applied to a rollout group $\{r_1, \dots, r_G\}$ where all r_i are incorrect is:

$$\nabla \mathcal{L}_{\text{GRPO}} \propto - \sum_{i=1}^G \nabla \log \pi_{\theta}(r_i | x)$$

which suppresses the probability of all sampled trajectories. Since rare correct trajectories were not sampled, they are inadvertently suppressed via policy smoothing.

Learning or Inference Procedure

The Per-Problem Base Anchoring (PBA) fix proceeds as follows:

1. Classify each training problem as boundary or safe using the base model’s entropy $H(\pi_{\text{base}}(\cdot | x))$; problems above a threshold H^* are classified as boundary.
2. For safe problems, apply standard RLVR with no modification.
3. For boundary problems, augment the rollout group with forced correct samples from the frozen base model before computing the RLVR gradient.
4. Additionally, apply a KL penalty on boundary problems:

$$\mathcal{L}_{\text{PBA}} = \mathcal{L}_{\text{RLVR}} + \lambda D_{\text{KL}}(\pi_{\text{base}} \| \pi_{\theta})$$

which anchors the policy toward the base distribution on these inputs.

What the Guarantee Says

PBA does not guarantee recovery of all lost pass@ k on boundary prompts. Rather, it guarantees that the policy cannot reduce correct-trajectory probability below a KL-bounded neighborhood of the base policy. For $\lambda > 0$ and $D_{\text{KL}}(\pi_{\text{base}} \| \pi_{\theta}) \leq \epsilon_{\lambda}$, the trained policy satisfies:

$$\pi_{\text{RL,correct}}(x) \geq \pi_{\text{base,correct}}(x) \cdot e^{-\epsilon_{\lambda}}$$

providing a fractional lower bound on retained diversity.

Experimental Findings

Experiments use Llama-3 8B and Qwen2.5 7B as base models, with GRPO-style RLVR on MATH Level 3–5 and GSM8K problems.

Model	Metric	Base	+RLVR
Qwen2.5-7B	pass@1	56.3	68.1
Qwen2.5-7B	pass@16	83.2	78.4
Llama-3-8B	pass@1	48.9	57.3
Llama-3-8B	pass@16	79.1	73.6

The pass@16 regressions are concentrated on boundary prompts (15–25% of the test set) and are negligible on safe prompts. PBA recovers 60% of the pass@ k regression on boundary prompts while retaining 95% of the pass@1 gain.

Ablations and Interpretation

Group size G : Larger rollout groups ($G = 16$ vs. $G = 8$) reduce but do not eliminate inversion, as boundary prompts can still yield zero correct rollouts when $\pi_{\text{base,correct}}(x) < 1/G$.

Problem difficulty: Pass@ k inversion is uncorrelated with pass@1 improvement magnitude; problems that yield large pass@1 gains may or may not be on boundary prompts.

Anchor weight λ : Higher λ recovers more pass@ k but reduces pass@1 gains. The optimal λ is task-dependent.

Broader implication: Evaluating reasoning models solely at pass@1 is insufficient for any deployment context using sampling-based inference. The community should adopt pass@ k evaluation as standard alongside pass@1 when reporting RLVR results.

Reference

Todd Zhou. **When RLVR Shrinks the Reasoning Boundary: Diagnosing Pass@ k Inversion.** arXiv:2607.20543, July 2026. <https://arxiv.org/abs/2607.20543>